

KAILEY SIMONS

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

Candidate for M.Eng. in Computer Science, Data Science, Economics

May 2027

Candidate for B.S. in Computer Science, Data Science, and Economics, and B.S in Business Analytics

May 2026

Relevant Coursework: Statistical Learning Theory, Inference, Design and Analysis of Algorithms, Econometrics

Data Science, Statistics, Optimization in Business Analytics, Machine Learning, Programming in Python

SKILLS & ACTIVITIES

Programming & Databases: Python, PostgreSQL, R, Julia, Java, TypeScript, HTML, STATA

Tools: AWS, Pandas, NumPy, Git, XGBoost, PyTorch, Scikit-learn, TensorFlow, Claude Code, JIRA, Confluence

Expertise: Data Science, Machine Learning (ML), Quantitative Analysis, Statistical Analysis, Research

Activities: MIT Varsity Swimming and Diving (All-American)

APPLIED ML RESEARCH EXPERIENCE

MIT Computer Science and Artificial Intelligence Laboratory

February 2026-Present

Applied ML Research Scientist

Cambridge, MA

- Analyzed ML models for Brazilian Portuguese audio deepfake detection; used RMS normalization to recover accuracy from 42% to 99.5% under codec compression (accepted to BRACIS; published in Springer Nature)
- Developed a framework for detecting and mitigating hallucinations in multi-agent financial QA systems (published in MDPI)

MIT Center for Collective Intelligence

May 2024-June 2025

Undergraduate Researcher

Cambridge, MA

- Implemented A/B testing and statistical tests (ANOVA, regression) to evaluate human-AI interactions

PROFESSIONAL EXPERIENCE

Accenture

June 2026-August 2026

Data Scientist

New York City, NY

- Led agentic AI rollout strategy, architecting 3 AI agents projected to each save ~20 analyst hours per week
- Spearheaded development of 8+ Copilot adoption analytics, presenting insights directly to client executives

Balyasny Asset Management - MIT Generative AI Lab

February 2026-May 2026

AI Engineer

Cambridge, MA

- Built and tested MVPs of a multi-agent generation and evaluation pipeline to summarize Power & Gas markets, improving report quality by 59.4% and reducing the number of contradictions in reports by 73.3% on average

Nasdaq

January 2026-February 2026

Data Science & AI Engineer

Boston, MA

- Evaluated and implemented two models from literature to improve the performance of an ML-powered system to track National Best Bid and Offer (NBBO) in a fast-paced trading environment to inform trading decisions

Vanguard

May 2025-August 2025

Investment Systems Software Engineer

Malvern, PA

- Spearheaded the integration of AI in the Portfolio Review Department's systems by designing an analytics system using AWS Bedrock, Lambda, and PostgreSQL, reducing analyst research time by 20%
- Developed a volatility nowcasting model and conducted exploratory data analysis in Python

Daylit

January 2025-February 2025

AI Engineer

Boston, MA

- Engineered and deployed a loan review pipeline using the Google BERT model for Retrieval Augmented Generation (RAG) in Python, reducing manual review time by 50%+ and saving 300+ analyst hours per quarter

LEADERSHIP EXPERIENCE

MIT Undergraduate Economics Association

September 2024-May 2026

Co-President (2025-2026)/Publicity Chair (2024-2025)

Cambridge, MA

- Directed event strategy for 100+ students, increasing event participation by 10%

MIT Undergraduate Association Sustainability Committee

September 2023-May 2026

Co-President (2025-2026)/Publicity Lead (2023-2025)

Cambridge, MA

- Led weekly meetings to coordinate and host events with 5+ organizations across campus or from other schools

PROJECTS

Paid Family Leave and Maternal Labor Supply Analysis

February 2026-May 2026

- Modeled the effect of paid family leave on maternal labor supply across industries by gender composition

Exploring Effectiveness of Using LLM-Based Chatbots for Combatting Misinformation

December 2025

- Led an experimental study on belief change, using Python for data cleaning, analysis, and regressions